

Order, Ranking and Sitting Arrangement

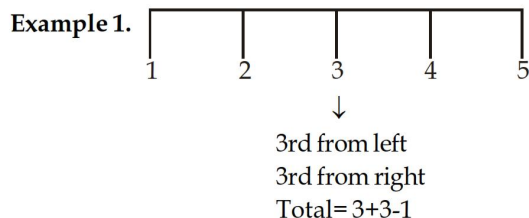
Order & Ranking Concepts:-

In this topic, generally the ranks of a person from both the sides left or right or from top and bottom are mentioned and total numbers of a persons are asked. sometime the questions are based on their interchanged positions.

Formulas to determine the positioning of a person

- (1) $\text{Left} + \text{Right} = \text{Total} + 1$
- (2) $\text{Left} = \text{Total} + 1 - \text{Right}$
- (3) $\text{Right} = \text{Total} + 1 - \text{Left}$
- (4) $\text{Total} = \text{left} + \text{Right} - 1$

Note : the above formulas are only for a single person's position



Same for vertical & Horizontal

- (1) $\text{Total} + 1 = \text{top} + \text{Bottom}$
- (2) $\text{Top} = \text{Total} + 1 - \text{Bottom}$
- (3) $\text{Bottom} = \text{Total} + 1 - \text{Top}$
- (4) $\text{Total} = \text{Top} + \text{Bottom} - 1$

SEATING ARRANGEMENT

✦ In seating arrangement, we are generally asked to arrange a group of people according to the given conditions. They may have to be seated around a table, the table could be of any shape-circular, square, rectangular, pentagonal or any other. To solve seating arrangement problems on the basis of the information given in the equation.

✦ These type of questions judge the ability of the candidates to analyze the information and solve the questions by the help of pictorial figures.

- **Linear arrangement:** - In this arrangement, there can be single row or parallel rows facing each other or opposite.

(i) $\text{Left} \longleftrightarrow \text{Right}$
(Directions if the people are facing north)

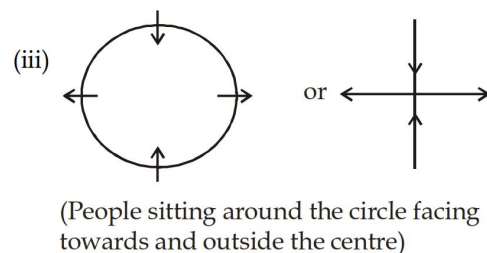
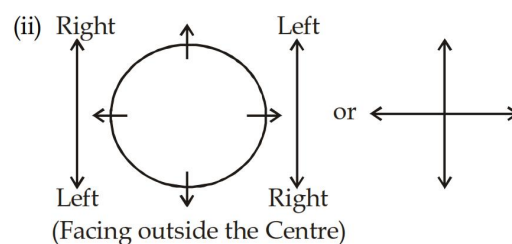
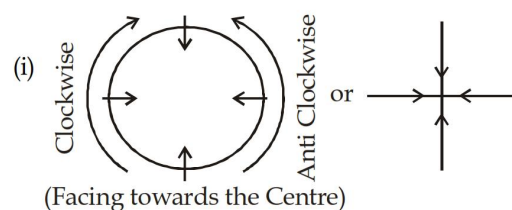
(ii) $\text{Right} \longleftrightarrow \text{Left}$
(Directions if the people are facing South)

(iii) $\text{Right} \longleftrightarrow \text{Left}$

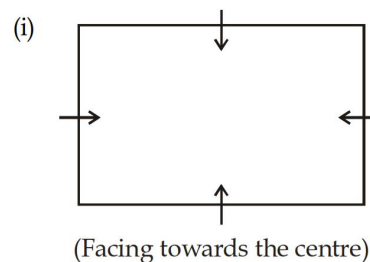
$\text{Left} \longleftrightarrow \text{Right}$

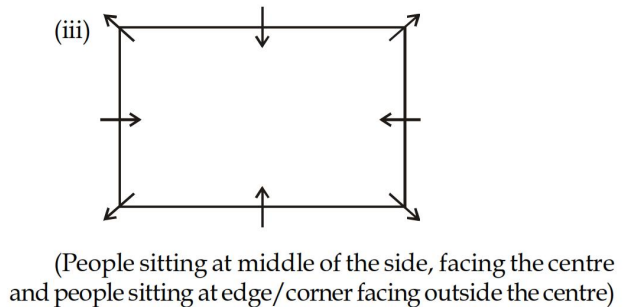
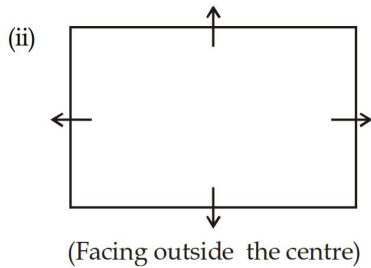
(Directions if the people are seating parallel to each other facing South and North)

- **Circular Arrangement:** In this Arrangement, people are sitting around a circle facing towards or outside the centre.



- **Rectangular/square Arrangement:** In this Arrangement, people are sitting around a rectangle or square facing towards or outside the centre.



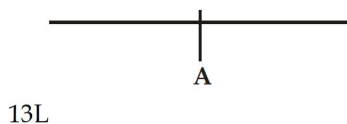


Solved Examples

1. In a row of 40 students, A is 13th from the left end, find the rank from right end.

- (a) 29 (b) 28
(c) 30 (d) 31

Sol. Total = 40

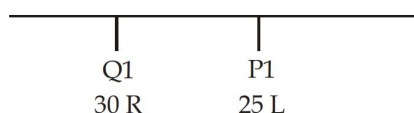


A's rank from right side = Total + 1 - left
= $40 - 13 + 1 = 27 + 1 = 28$

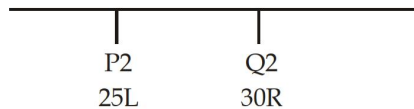
2. In a row 'P' is 25th from left end, Q is 30th from right end. Find the total no. of students in all.

Sol. Can't be Determined as there are more than 1 possibilities

Case 1



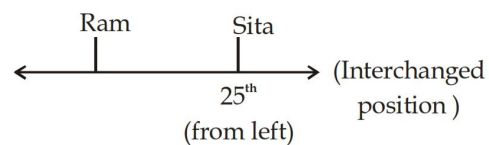
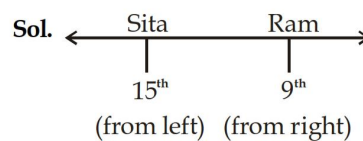
Case 2



Note: When total is not given and 2 persons positions from left and right are given, then answer is Cannot be determined

3. In a row, Ram is 9th from the right. Sita is 15th from the left. If Ram and Sita interchange their positions, Sita will be 25th from left. Which of the following will be Ram's position from the right?

- (a) 17 (b) 18
(c) 19 (d) 20



Total = $(25 + 9) - 1 = 34 - 1 = 33$

So, Ram's position from the right = $(33 - 15) + 1$
= $18 + 1 = 19$

4. In any row Zafar is 15th from top and Jitender is 11th from bottom. 23 persons are present between them, then find out the total no. of students in row.

- (a) 48 (b) 49
(c) 50 (d) 51

Sol. Total number of students = Zafar's position from top + Jitender's position from bottom + persons present between them
= $15 + 11 + 23 = 49$

5. Umesh is taller than Satish, Suresh is shorter than Neeraj but taller than Umesh, who is the tallest among them?

- (a) Umesh (b) Suresh
(c) Satish (d) Neeraj

Sol. Umesh > Satish

Neeraj > Suresh > Umesh

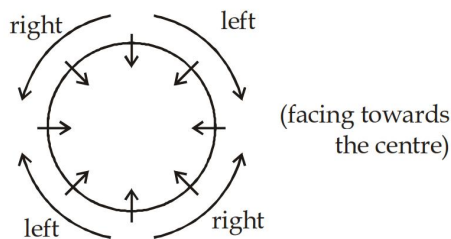
∴ Neeraj > Suresh > Umesh > Satish

Clearly, Neeraj is tallest among them.

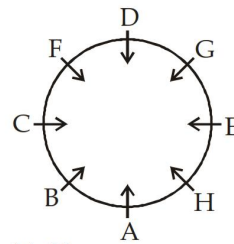
Directions (6 - 9): Read the information carefully and answer the questions given below:

A, B, C, D, E, F, G and H are the eight friends, they are sitting in a circle facing center. A is second to the right of C and fourth to the left of D, E is not immediate neighbour of C and D. H is next to the right of A. F is second to the right of G. G is not immediate neighbour of C.

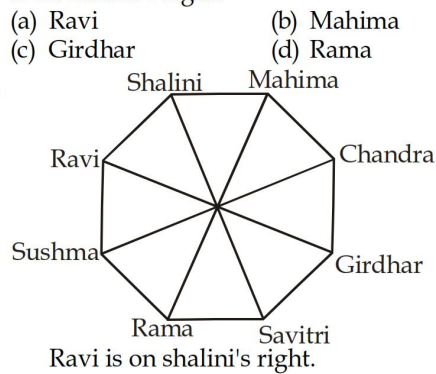
6. Who is second to the left of G?
 (a) H (b) D
 (c) A (d) E
7. Who is fourth to the right of C ?
 (a) B (b) A
 (c) H (d) E
8. In which pair second person is just right to first?
 (a) HE (b) FB
 (c) GF (d) CA
9. Who is the immediate neighbour of H from both side?
 (a) AG (b) EG
 (c) AE (d) BC
- Sol.** When eight persons are sitting in a circle we arrange them in following way.



It should be kept in mind that left and right sides are always opposite, those facing each other in a circle.
 According to the question, sitting arrangements of eight friends are shown in the figure given below:-



6. (a); H 7. (d); E
 8. (a); HE 9. (c); AE
10. A group of friends are sitting in an arrangement one each at the corner of an octagon. All are facing the centre. Mahima is sitting diagonally opposite to Rama, who is on sushma's right. Ravi is next to sushma and opposite to Girdhar, who is on chandra's left. Savitri is not on Mahima's right but opposite to shalini. Who is on shalini's right?



Practice Set

- In a class of 45 student Aditya's rank is twelve from top what is his rank from bottom.
 (a) 33 (b) 34
 (c) 35 (d) None of these
- In a class of 42 students Nutan's Rank is 22 from bottom. What is her rank from top?
 (a) 21 (b) 22
 (c) 23 (d) cannot be determined
- In a class, Sonal rank is 10th from top. What is his rank from below?
 (a) 24 (b) 25
 (c) 26 (d) cannot be determined
- In a class Sonu's rank is 15th from top and twelve from bottom. How many students are there in that class?
 (a) 21 (b) 25
 (c) 26 (d) cannot be determine
- In a class of 80 boy a student rank is 48th from top. What is his rank from bottom?
 (a) 32 (b) 34
 (c) 35 (d) 33
- In a class, Abhinav's rank is 79 from bottom while there are 80 boy in class what is Abhinav rank from top?
 (a) 2 (b) 3
 (c) 4 (d) cannot be determine
- In a class of 41 children, Aditya's rank is eight from top. Mamta is seven ranks below Aditya. What is Mamta rank from bottom?
 (a) 26 (b) 27
 (c) 25 (d) Cannot be determine
- In a class of 34 children, Ajay's rank from the top is twelve. Manoj is eight ranks below Ajay. What is Manoj's rank from the bottom?
 (a) 15 (b) 16
 (c) 14 (d) cannot be determine
- In a row of boys, Aditya is seventh from the start and eleventh from the end. In another row of girl, Nutan is tenth from the start and twelfth from the end. How many student are there in both the rows together?

- (a) 36 (b) 37
(c) 39 (d) None of these
10. In a row of girls, Priya is thirteenth from the left and Dauli is seventeenth from the right. If in this row Priya is eleventh from the right then what is the position of Dauli from the left?
(a) 6th (b) 7th
(c) 10th (d) 12th
11. Sonu is fifteenth from the front in a column of boys. There were thrice as many behind him as there were in front. How many boys are there between Sonu and the seventh boy from the end of the column?
(a) 33 (b) 34
(c) 35 (d) Data inadequate
12. Forty boys are standing in a row facing the north. Amit is eleventh from the left and Sanjay is thirty-first from the right end of the row. How far will Shreya, who is third to the right of Amit in the row, be from Sanjay?
(a) 2nd (b) 3rd
(c) 4th (d) 5th
13. In a class, among the passed students, Amisha is twenty-second from the top and Anuja, who is 5 ranks below Amisha, is thirty-fourth from the bottom. All the students from the class have appeared for the exam. If the ratio of the students who passed in the exam to those who failed is 4 : 1 in that class, how many students are there in the class?
(a) 60 (b) 75
(c) 90 (d) Data inadequate
14. In a row of girls facing North, Soni is 10th to the left of Pallavi, who is 21st from the right end. If Malini, who is 17th from the left end, is fourth to the right of Soni, how many girls are there in the row?
(a) 37 (b) 43
(c) 44 (d) Data inadequate
15. In a row of 40 boys, Aditya was shifted 10 places to the right of Rohan and Kewal was shifted 10 places to the left of Vilas. If Vilas was twenty-sixth from the left and there were three boys between Kewal and Aditya after shifting, what was the position of Rohan in the row?
(a) 10th from the right end
(b) 10th from the left end
(c) 39th from the right end
(d) Data inadequate
16. Raj is seventeenth from the left end of a row of 29 boys and Karan is seventeenth from the right end in the same row. How many boys are there between them in the row?
(a) 3 (b) 5
(c) 6 (d) Data inadequate
17. In a row of forty children, A is thirteenth from the left end and Q is ninth from the right end. How many children are there between A and R if R is fourth to the left of Q?
(a) 12 (b) 13
(c) 14 (d) 15
18. In a row of girls, there are 16 girls between Priya and Natasha. Priya is thirty-second from the left end of the row. If Priya is nearer than Natasha to the right end of the row, then how far away is Natasha from the left end of the row?
(a) 16th (b) 14th
(c) 15th (d) Data inadequate
19. George is fifth from the left and Peter is twelfth from the right end in a row of children. If Peter shifts by three places towards George, he becomes tenth from the left end. How many children are there in the row?
(a) 21 (b) 22
(c) 23 (d) 24
20. In a row of boys, if A who is tenth from the left and B who is ninth from the right interchange their positions, A becomes fifteenth from the left. How many boys are there in the row?
(a) 23 (b) 27
(c) 28 (d) 31
21. In a class, the ratio of boys and girl is 3 : 2. Neelam's rank from bottom is fourteenth and sixth rank from girls. If Neelam's rank 32th from top then how many girls are there above from Neelam in class?
(a) 13 (b) 14
(c) 12 (d) 11
22. In a row of students. 'A' is 10th from the top. In the same row B is 25th from the bottom and 27th from the top. Then what will be A's position from the bottom?
(a) 44 (b) 43
(c) 42 (d) 41
23. In row, X is 30th position from the left. Y is 27th position from the right. If there are 40 students in the line then how many students are there between X and Y?
(a) 17 (b) 15
(c) 16 (d) Data inadequate
24. A cricket ball is lighter than a hockey ball. A volleyball is lighter than football. The hockey ball is lighter than the football but heavier than a tennis ball. Which of the following is the heaviest?
(a) Hockey ball (b) Football
(c) Cricket ball (d) Volley ball
- Directions (25 - 30) :** Eight persons are sitting in two parallel lines. In each parallel lines four persons are sitting at equal distance. Line I: P, Q, R, S are sitting. (not necessary, they are sitting in series given above) And they

are facing towards north. Line II, A, B, C, D are sitting (not necessary they are sitting in a series given above) they are facing south. They are all sitting in such a way that line I and line II persons facing to each other.

B is second to the left of D. R is sitting to next who is facing D. Only one person is sitting between R and P. C. is not facing to R. Two persons are sitting between R and Q.

25. Who is facing to P?

- (a) A (b) B
(c) C (d) D

26. Which of the following is sitting to next to the left who is facing D?

- (a) P (b) Q
(c) R (d) S

27. Which of the following is sitting to next to the left who is facing B?

- (a) P (b) Q
(c) R (d) S

28. Which of the following statement is true about S?

- (a) S is the last person in a line
(b) S is second to the right of Q.
(c) S is not an immediate neighbour of P
(d) None of these

29. Four of the following five are alike in a certain way and makes a group. Which is one that does not belong to that group?

- (a) DS (b) PB
(c) QB (d) RA

30. A, B, C, D, E, F and G are sitting in a row facing north. F is sitting next to the right of E. C is the neighbour of B and D. The person who is sitting third to the left of D is Ist of an end of row, E is fourth to the right of G. Then which position for A is?

- (a) between C and E (b) Ist from right end
(c) Ist from left end (d) Data inadequate

Directions (31 - 35) : Read the following information carefully and answer the question given below it:

A, B, C, D, E, F, G and H are sitting in a circle facing centre. C is third to the left of A and second to right of E. B is second to the right of C. D is second to the right of F, who is second to the right of A. G is not an immediate neighbour of C.

31. Who is immediate right of C?

- (a) H (b) G
(c) D (d) Data inadequate

32. Who is immediate right of H?

- (a) D (b) G
(c) A (d) None of these

33. Who is immediate left of D?

- (a) C (b) H
(c) F (d) E

34. Who is third to the right of D?

- (a) C (b) H
(c) E (d) None of these

35. Who is second to the right of G?

- (a) A (b) D
(c) E (d) B

Distinct Questions

Directions (36 - 43) : Read the following informations carefully and answer the questions given below it:

Eight family members A, B, C, D, E, F, G and H are sitting in a circle facing centre. 'F' the wife of 'D' is sitting third to the right of C. 'A' is the son of H. A is second to the left of D. D is not an immediate neighbour of F or C. Also no male family member is immediate neighbour of D. 'G' is second to the left of the son of D. Between H and the brother of A only two persons are sitting. C and D are not the brother of A. The son of 'D' and the wife of D are not sitting immediate. 'F' is the mother of H. F is not an immediate neighbour of B and 'G'. G is the sister of E. C is wife of D's son.

36. Who is second to the left of G?

- (a) Brother of A (b) Mother of A
(c) D (d) Mother of B

37. Who is the son of D?

- (a) E (b) G
(c) A (d) B

38. How many persons are there between A and his brother?

- (a) None (b) One
(c) Two (d) Three

39. Who is sitting between H and F?

- (a) the wife of D (b) the son of D
(c) C (d) A

40. Who is the brother of 'A'?

- (a) E (b) G
(c) A (d) B

41. Based on the given information above, what is the relation of A to D?

- (a) Grandfather (b) Son
(c) Grandson (d) Wife

42. Four of the following five are alike in a certain way based on the information given above which is one that does not belong to that group?

- (a) B (b) C
(c) H (d) G

43. Which of the following information given below about 'H' is definite true.

- (a) H is a man (b) H is cousin of C
(c) H is the wife of D
(d) To left and right of H, immediate sitting members both are man.

44. In a straight line, some boys are standing facing north. A is third from the right end. B is second from the left end. C is fourth to the left of A. B is sixth to the left of C. Then how many boys are there in the line?
- (a) 13 (b) 14
(c) 15 (d) 16
45. In a row, there are 50 students standing facing towards north. 'X' is the 5th from right end. 'Y' is 9th from the left end. Then how many students are there between X and Y?
- (a) 38 (b) 34
(c) 36 (d) 33
(e) None of these
- Direction (46 to 50) :** The questions are based on the information given below:
- A, B, C, D, E, F, G and H are sitting around a round table facing centre. There are four male person and four female person in the group. But no two male and no two female are immediate neighbour.
- A is the wife of H and she is sitting third to the left of E. F is second to the right of D. D is not immediate neighbour of A and E. H is sitting beside C. F and his wife B are not sitting beside.
46. What will be true about 'G' in the given option?
- (a) G is a male person
(b) G is sitting between 'F' and 'H'
(c) G is third to the left of E
(d) G is second to the right of B
47. Who is third to the left of B?
- (a) F (b) H
(c) D (d) A
48. How many persons are there between B and F if they want to move anti-clockwise direction from B?
- (a) One (b) Two
(c) Three (d) Four
49. Four of the following five are alike in a certain way and makes a group. Which is one that does not belong to that group?
- (a) H (b) F
(c) E (d) G
50. Which of the following options is related to only female person?
- (a) A, B, H (b) G, F, C
(c) C, H, G (d) None of these

Previous year Questions (Memory Based)

1. There are five friends- Satish, Kishore, Mohan, Anil and Rajesh. Mohan is the tallest. Satish is shorter than Kishore but taller than Rajesh. Anil is little shorter than Kishore but little taller than Satish. Who is taller than Rajesh but shorter than Anil?
- (a) Anil (b) Kishore
(c) Rajesh (d) Satish
2. Banu is twice as old as Anehu but twice younger than Fathima. Caroline is half the age of Anehu but is twice order than Dayana. Which two persons from the pair of the oldest and youngest ?
- (a) Banu and Dayana
(b) Fathima and Anehu
(c) Fathima and Dayana
(d) Banu and Caroline
3. Raju walks slower than Raghu and Raghu walks as fast as Guru and Krishna walk faster than Guru. Who walks the fastest?
- (a) Raghu (b) Raju
(c) Krishna (d) Both Raghu and Raju
4. In a class, Rajan got the 11th rank and he was 31st from the bottom of the boys passed. Three boys did not take the examination and one failed. What is the total strength of the class?
- (a) 35 (b) 42
(c) 45 (d) 46
5. In a row of 16 boys, when Prakash was shifted by two place towards the left, he became 7th from the left end. What was his earlier position from right end of the row?
- (a) 7th (b) 8th
(c) 9th (d) 10th
6. In a row of children, A is 9th from the left and B is fifth from right. When A and B interchange their position A will be 18th from the left. What will be B's position from the right?
- (a) 9 (b) 5
(c) 18 (d) 14
7. In a row, Ram is 7th from the left and Sita is 5th from the right. When they interchange their position then Sita will be 19th position from the right. What will be Ram's position from the left?
- (a) 20 (b) 21
(c) 22 (d) 23
8. In a row, Ram rank is 18th from the top. Sita is five rank below Ram. Laxman is 19th from the bottom. Sita is sitting equal distance from Ram and Laxman. Then how many children are there in row?
- (a) 45 (b) 44
(c) 43 (d) 46
9. Sohan ranks fourteenth from top in a class and twelfth from bottom. How many students are there in the class?

- (a) 25 (b) 26
(c) 24 (d) None of these
10. Mohan ranks thirteenth in a class of twenty students. What is his rank from the bottom?
(a) 7 (b) 8
(c) 6 (d) 9
11. In a queue, Sadiq is 14th from the front and Joseph is 17th from the end, while Jane is in between Sadiq and Joseph. If Sadiq be ahead of Joseph and there be 48 persons in the queue, how many persons are there between Sadiq and Jane?
(a) 5 (b) 6
(c) 7 (d) 8
12. In a row of children Mihir is third to right of Sunil, who is ninth from the left end of the row. What is Sunil's position from the right end of the row?
(a) Twelfth (b) Thirteenth
(c) Fourteenth (d) Data inadequate
13. In a class of 40 children, Surender's rank is eight from the top. Sujit is five ranks below Surender. What is the Sujit's rank from bottom?
(a) 27 (b) 29
(c) 28 (d) 26
14. In the class of 35 students, Shyam is fifteenth position from top. According to series five students are below Amit. How many students are there between Amit and Shyam?
(a) 20 (b) 15
(c) 14 (d) 12
15. In a row of children, Neeta is fifteenth from the left end of the row. If she shifted towards the right end of the row by four places, she becomes eighth from the right end. How many children are there in the row?
(a) 27 (b) 26
(c) 28 (d) 24
16. Mahesh is 5th from left and Rakesh is 7th from right in a row of children. If they both interchange their position then Rakesh will be 25th from right end of the row. Then how many children are there between Rakesh and Mahesh?
(a) 19 (b) 22
(c) 17 (d) 25
17. In a row, Vishesh is 11th from the left and Garima is 15th from right, If they both interchange their position then Vishesh will be 15th from left then what will be Garima's position from right?
(a) 17 (b) 18
(c) 19 (d) 20

Directions (18 - 22): Read the information given below and then answer the questions that follow:-

Eight friends M, N, O, P, Q, R, S and T are sitting in a circle facing the centre. P is third to the right of M and second to the left of S. T is third to the left of S. Q is not immediate neighbour of S. N is not immediate neighbour of M. O is second to the right of N. Q is also not immediate neighbour of T.

18. Who is third to the right of N?
(a) Q (b) T
(c) P (d) R
19. Who is second to the left of M?
(a) N (b) R
(c) S (d) None of these
20. Who is sitting between P and S?
(a) Only R (b) R and N
(c) Only N (d) None of these
21. How many persons are sitting between Q and S?
(a) Only 2 (b) Only 3
(c) Only 4 (d) None of these
22. Who is sitting between T and M?
(a) Only R (b) Only N
(c) Q and P (d) None of these
23. Six persons A, B, C, D, E and F are sitting in a circle facing centre. B is sitting between F and C. A is sitting between E and D. F is sitting to the left of D. Who is sitting between A and F?
(a) B (b) C
(c) E (d) D
24. Five persons P, Q, R, S and T are standing on a circle, facing centre. Q is standing between T and S. P is not nearest left of S. Then who is second to the right of Q?
(a) P (b) R
(c) S (d) Data inadequate
25. Five members of an interview committee are sitting in a line. B is sitting to the left of E and to the right of A. D is sitting to the right of C but to the left of A. Who is sitting in the middle of the line?
(a) B (b) A
(c) D (d) C
26. Four children, P, Q, R and S are sitting on a ladder. P is sitting above from Q. Q is sitting between P and R. If S is sitting above P then who is sitting third from the bottom?
(a) Q (b) R
(c) P (d) S
27. An accused is more powerful than the deceased. Police are less powerful than the court but more powerful than the lawyer. Accused bends his head before police. Who is the most powerful?
(a) Court (b) Accused
(c) Police (d) Lawyer

28. X is mightier than Y and Y is mightier than Z. P is mightier than Q but inferior to Y. Q is mightier than Z. Who is the weakest among all ?
 (a) X (b) Z
 (c) Q (d) P
29. Few students are sitting in a row. In that row Ram is 15th position from left and 23rd position from right. Find out the total no. of students in row?
 (a) 38 (b) 37
 (c) 40 (d) 36
30. In a row, 50 girls are sitting. Sita's position is 27th from left then find out the position of Sita from right?
 (a) 22 (b) 23
 (c) 24 (d) 25
31. Saran is eighteenth from the right end in a row of 50 boys. What is his position from the left end?
 (a) 32 (b) 35
 (c) 33 (d) 34
32. In a class of 90, where girls are twice that of boys, Shridar ranked fourteenth from the top, if there are 10 girls ahead of Shridar, how many boys are after him in rank?
 (a) 23 (b) 26
 (c) 25 (d) 22
 (e) None of these
33. Sita ranks nineteenth in a class of 68 students. What is her rank from last?
 (a) 50 (b) 51
 (c) 49 (d) 48
34. Raji is 5 rank ahead of Raj in a class of 46 students. If Raj's rank is twelfth from the last, what is Raji's rank from the start?
 (a) 29 (b) 31
 (c) 28 (d) 30
35. Karthick is 6 ranks ahead of Subash who ranks sixteenth in a class of 42. What is karthick's rank from the last?
 (a) 33 (b) 32
 (c) 31 (d) 30
36. A ranks fourth in a class. B ranks ninth from the last. If C is ninth after A and just in the middle of A and B, How many students are there in the class?
 (a) 33 (b) 32
 (c) 31 (d) 30
37. Akhil ranked seventeenth from the top and thirty seventh from the bottom in a class. How many students are there in the class?
 (a) 53 (b) 45
 (c) 54 (d) 52
38. Shakthi ranks eleventh in a class of 54 students. What is his rank from last?
 (a) 43 (b) 44
 (c) 42 (d) 40
39. Naresh is 22nd from left end in a row of 47 boys. What is his position from right end?
 (a) 24 (b) 25
 (c) 23 (d) 26
40. Reshma and Praveena are ranked ninth and thirteenth from the top in a class of 57 students. What will be the respective ranks from the bottom of the class?
 (a) 48, 44 (b) 49, 45
 (c) 45, 49 (d) 47, 43

Practice Set Solutions

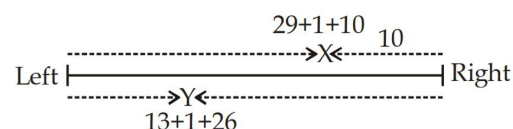
1. (b); rank from bottom [(total no. of students + 1) - rank from top] = [(45 + 1) - 12] = 34th
2. (a); (42 + 1) - 22 = 21
3. (d); cannot be determined
4. (c); total no. of student = [(rank from top + rank from bottom) - 1] = 15 + 12 - 1 = 26
5. (d); [(80 + 1) - 48] = 81 - 48 = 33
6. (a); [(80 + 1) - 79] = 2
7. (b); Mamta Rank from top = 8 + 7 = 15
 From below her rank = (41 + 1 - 15) = 27
8. (a); Ajay rank from top = 12 + 8 = 20
 Ajay rank from bottom = (34 + 1 - 20) = 35 - 20 = 15
9. (d); in first row = (7 + 11 - 1) = 17 students
 In second row = (10 + 12 - 1) = 21 students
 Total = 21 + 17 = 38 students
10. (b); Clearly, Priya is 13th from the left and 11th from the right end of the row. So, number of girls in the row = (13 + 11 - 1) = 23. Now, Dauli is 17th from the right. Number of girls to the left of Dauli = [(23 + 1) - 17] = 7. Hence, Dauli is 7th from the left end of the row.
11. (c); Number of boys in front of Sonu = 14. Number of boys behind Sonu = (14 * 3) = 42. So, Total number of boys in the column = (14 + 1 + 42) = 57. In a column of 57 boys, the seventh boy from the end is clearly 51st from the start. Thus, we have to find the number of boys between the 15th and the 51st boy, which is clearly 35.
12. (c); Number of boys to the left of Sanjay = (40 - 31) = 9. So, Sanjay is 10th from the left end. Shreya is third to the right of Amit. So, Shreya is 14th from the left end. Clearly, Shreya is fourth to the right of Sanjay.
13. (b); Amisha is 22nd from the top and Anuja is 5 ranks below Amisha. So, Anuja is 27th from the top. Also, Anuja is 34th from the bottom. So, Number of students passed = (26 + 1 + 33) = 60; Let the number of students passed and the number failed be 4x and x respectively. Then, 4x = 60 or x = 15. Hence, number of students in the class = (60 + 15) = 75.

-
- Diagram illustrating the addition of two vectors P and N to find their resultant R . Vector P has a magnitude of 31, and vector N has a magnitude of 16. The resultant vector R is shown as the sum of P and N , with a magnitude of 31.

Number of girls to the left of Natasha in II = $[31 - (1 + 16) = 14]$. Clearly, Natasha is 15th from the left end of the row.

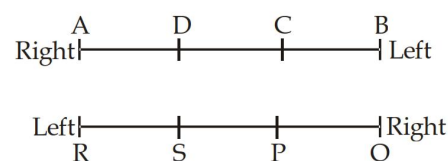
21. (c); Total number of students in class
 $= (14+32) - 1 = 46 - 1 = 45$
 $\therefore$ Ratio $\Rightarrow 2 : 3$
 $2x + 3x = 45$
 $5x = 45$
 $x = 9$

23. (b);

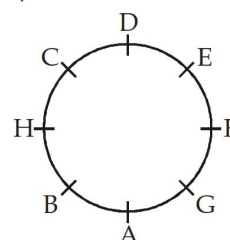


24. (b);

Directions (25 - 29):



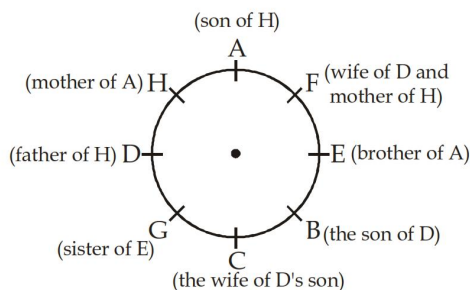
- Directions (31 - 35) :**



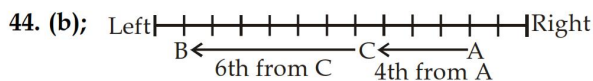
31. (a); H is immediate right of C.
32. (d); B is immediate right of H.
33. (d); E is the immediate left of D.
34. (d); B is the third to the right of D.
35. (c); E is second to the right of C.

Distinct Solutions

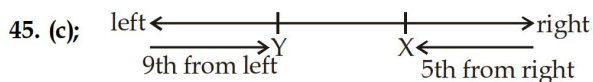
Directions (36 - 43):



36. (b); Mother of G is the second to left of G because A, E and G are brother-sister and H is mother of A.
 37. (d); B is the son of D.
 38. (b); Only one person is sitting between A and his brother.
 39. (d); A is sitting between F and H.
 40. (a); E is the brother of A.
 41. (c); A is the grandson of D.
 42. (a); All except 'B' are women.
 43. (d); To left and right of H immediate sitting both are men.



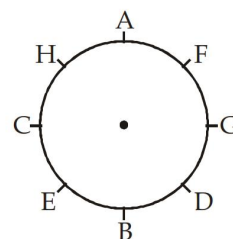
So, total number of boys are
 $= 1 + 6 + 4 + 3 = 14$



So clearly $= 50 - (9+5) = 50 - 14 = 36$

Direction (46 to 50):

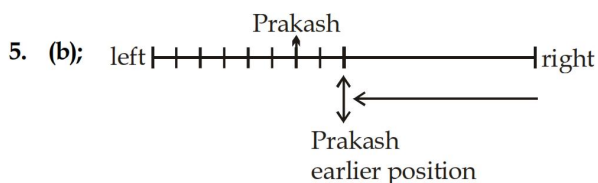
A	Female	wife of H
B	Female	wife of F
C	Female	
D	Male	
E	Male	
F	Male	husband of B
G	Female	
H	Male	husband of A



46. (d); G is second to the right of B.
 47. (b); H is third to the left of B.
 48. (b); Two
 49. (d); All except 'G' are male person.
 50. (d); None of these.

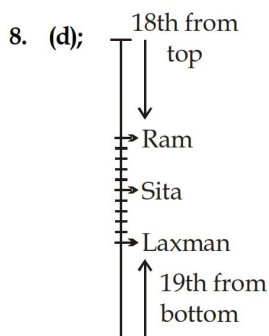
Previous year Solutions (Memory Based)

1. (d); Tallest $\Rightarrow$ Mohan
 Kishore > Satish > Rajesh
 Kishore > Anil > Satish
 So, Mohan > Kishore > Anil > Satish > Rajesh
 2. (c); Fatima > Banu > Anehu
 Anehu > Caroline > Dayana
 Fatima > Banu > Anehu > Caroline > Dayana
 3. (c); Raghu > Raja
 Kishna > Raghu = Gure
 So, clearly Krishna walks fastest.
 4. (c); Rajan rank from top = 11
 Rajan rank from bottom = 31
 Total students $= (11 + 31) - 1$
 $= (42 - 1) = 41$
 did not take the exam = 3
 failed = 1
 total = $41 + 3 + 1 = 45$



Prakash earlier position from left is 9th.
 Total no. of students is = 16

- $\therefore$ Prakash position from right $= (16-9)+1=7+1=8$ th
 6. (d); Total number of children $= (18+5)-1=23-1=22$
 So, B's position from right $= (22-9)+1=13+1=14$ th
 7. (b); Total number of boys $= (19+7) - 1 = 26 - 1 = 25$
 So, Ram position from left $= (25-5)+1=20+1=21$



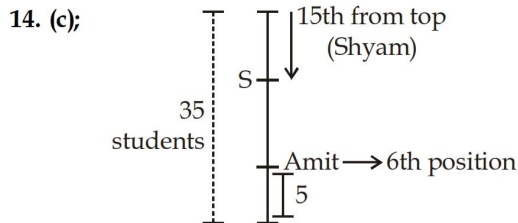
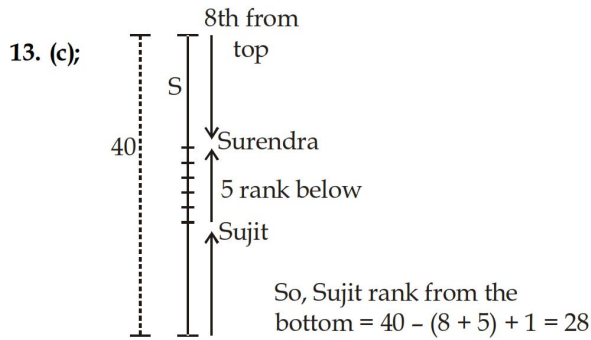
So, the total number of children $= 18 + 4 + 1 + 4 + 19 = 46$

9. (a); Student in the class $= (14+12)-1=26-1=25$
 10. (b); Mohan's rank from bottom $= (20-13)+1=7+1=8$ th

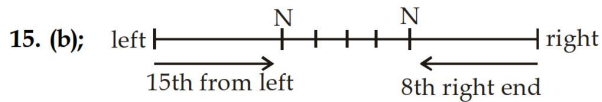
11. (d); Number of persons between Sadiq and Joseph
 $= 48 - (14+17) = 17$

As, Jane is in between Sadiq and Joseph.
 So, there are 8 persons between Sadiq and Jane.

12. (d); Data is inadequate



No of students $= 35 - (15+6) = 35 - 21 = 14$



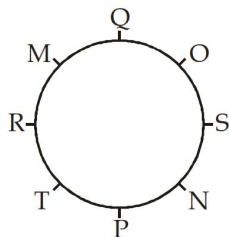
$(15+4+8) - 1 = 27 - 1 = 26$

So, clearly there are 26 children in the row.

16. (c); Total number of children $= (25+5) - 1 = 30 - 1 = 29$
 Children between Rakesh and Mahesh
 $= 29 - (\text{both sitting position})$
 $= 29 - (7+5)$
 $= 29 - 12 = 17$

17. (c); Total number of students $= (15+15) - 1 = (30-1) = 29$
 So, garima position from right $= (29-11) + 1$
 $= 18 + 1 = 19$

(18 - 22) :



18. (a); Q is third to the right of N.

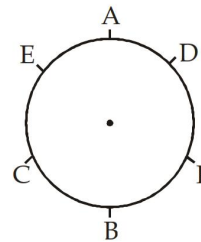
19. (d); O is second to left of M.

20. (c); Only 'N' is sitting between 'P' and 'S'.

21. (d); None of these.

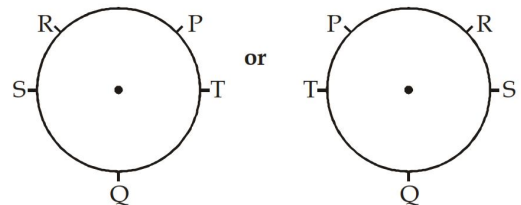
22. (a); 'R' is sitting between 'T' and 'M'.

23. (d);



$\therefore$ D is sitting between A and F.

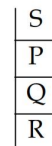
24. (d);



Above both two position are true. So, we can not define that who will be the second to the right of Q.

25. (b);
-
- clearly, in the middle of line A is sitting

26. (c);



So, P is third from bottom.

27. (a); accused > deceased

court > police > lawyer

police > accused

so, court > police > lawyer

> accused > deceased

so court is most powerful.

28. (b); $X > Y > Z$

$Y > P > Q$

$Q > Z$

so, $X > Y > P > Q > Z$

Z is the weakest among all.

29. (b); $T = 15 + 23 - 1$
 $= 37$
30. (c); $R = T - L + 1$
 $= 50 - 27 + 1$
 $= 24$
31. (c); $50 - 18 = 32$; Saran is 33 from the left.
32. (b); No of boys = x ; No. of girls = $2x$;
 $x + 2x = 90 \Rightarrow 3x = 90$
 x (Boys) = 30 ; $2x$ (Girls) = 60
 Number of student behind Shridar = $90 - 14 = 76$
 No of girls behind Shridar = $60 - 10 = 50$
 No of boys behind Shridar = $76 - 50 = 26$
33. (a); $68 - 19 = 49$, Hence 50th rank from last.
34. (d); No of students ahead of Raji in a rank = $46 - 17 = 29$
 Raji is 30th rank from the first
35. (a); Number of students behind Karthick = $42 - 10 = 32$
 Karthick rank 33rd from the last.
36. (d); ----A-----C-----B-----
 $3 + 1 + 8 + 1 + 8 + 1 + 8 = 30$
37. (a); $16 + 1 + 36 = 53$ students.
38. (b); $54 - 11 = 43$, Hence 44th rank from last.
39. (d); $47 - 22 = 25$, Hence 26th from the right end.
40. (b); Reshma rank = $57 - 9 = 48$, Reshma 49th from the bottom.
 Praveena rank = $57 - 13 = 44$, Praveena 45th from the bottom